

Lab 2 - Basic Networking Tools

Networking Tools

- IFCONFIG
- PING
- TRACEROUTE
- NETSTAT
- NMAP

Basic Linux Commands

- Change Directory: **CD <path_to_Directory>**
- Create New Directory: **mkdir <directory_name>**
- Create New File: **touch <file_name_with_extention>**
- Read File: **cat <file_name_with_extention>**
- Edit a file:
  - **gedit <file_name_with_extention>** (If gedit is installed) or
  - **code <file_name_with_extention>** (if vscode is installed)
- List files in a directory: **ls** or **ls <path_to_directory>**
- Read manual of a particular command: **man <command_name>**
- To get help for a particular command: **<command_name> --help** or **<command_name> -h**
- Tool installation: **sudo apt-get install <tool_name>**
- Run a bash script: **bash <script_name_with_extention_.sh>**

IFCONFIG / IPCONFIG

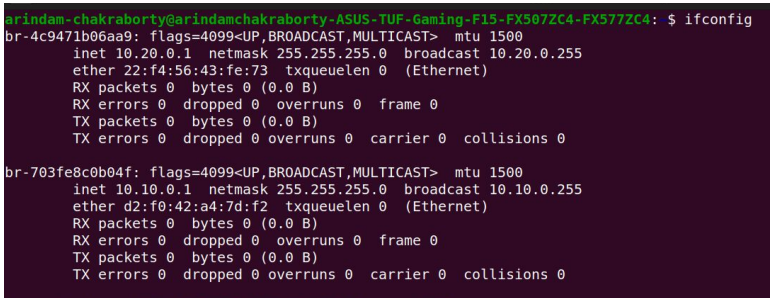
PURPOSE

- Used to configure, manage, and display network interface parameters in Unix-like OS.
- It provides information about network interfaces (e.g., IP addresses, MAC addresses, status, etc.) and allows users to enable or disable interfaces, set IP addresses, configure routes, etc.

USAGE: ifconfig [interface] [options]

Examples:

1. **ifconfig**: display all active network interfaces
2. **ifconfig eth0 down**: disables the network interface eth0



PING

PURPOSE : Used to check the network connectivity between two systems using ICMP protocol

USAGE : ping [options] destination

Example:

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ ping iitbhilai.ac.in
PING iitbhilai.ac.in (192.168.10.115) 56(84) bytes of data.
64 bytes from 192.168.10.115 (192.168.10.115): icmp_seq=1 ttl=62 time=0.462 ms
64 bytes from 192.168.10.115 (192.168.10.115): icmp_seq=2 ttl=62 time=0.645 ms
64 bytes from 192.168.10.115 (192.168.10.115): icmp_seq=3 ttl=62 time=0.701 ms
64 bytes from 192.168.10.115 (192.168.10.115): icmp_seq=4 ttl=62 time=0.701 ms
64 bytes from 192.168.10.115 (192.168.10.115): icmp_seq=5 ttl=62 time=0.698 ms
^C
--- iitbhilai.ac.in ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 0.462/0.641/0.701/0.092 ms
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```

TRACEROUTE / TRACERT

PURPOSE:

Used to trace the route that packets take from the source to the destination host. It shows the path, including the intermediate routers or hops, and measures the time taken for each hop. It helps in diagnosing network delays, locating bottlenecks, or identifying routing issues.

USAGE: traceroute [options] destination

Example:

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ traceroute iitbhilai.ac.in
traceroute to iitbhilai.ac.in (192.168.10.115), 30 hops max, 60 byte packets
 1  _gateway (10.30.0.1)  0.514 ms  0.489 ms  0.469 ms
 2  10.200.10.14 (10.200.10.14)  0.302 ms  0.284 ms  0.622 ms
 3  192.168.10.115 (192.168.10.115)  0.776 ms *  0.677 ms
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```

NETSTAT

PURPOSE

Used to display network connections, routing tables, interface statistics, and other network-related information on a system.

USAGE: netstat [options]

EXAMPLE:

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4: $ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 arindamchakrabort:55892 whatsapp-cd:xmpp-client ESTABLISHED
tcp        0      0 arindamchakrabort:51716 pnbomb-ad-in-f4.1:https CLOSE_WAIT
tcp       25      0 10.10.7.159:36226       10.10.1.46:ipp          CLOSE_WAIT
tcp       25      0 10.10.7.159:50224       10.10.1.46:ipp          CLOSE_WAIT
tcp       25      0 10.10.7.159:37204       10.10.1.46:ipp          CLOSE_WAIT
tcp       25      0 10.10.7.159:37280       10.10.1.46:ipp          CLOSE_WAIT
tcp       25      0 10.10.7.159:45044       10.10.1.46:ipp          CLOSE_WAIT
tcp       25      0 10.10.7.159:34106       10.10.1.46:ipp          CLOSE_WAIT
tcp       25      0 10.10.7.159:44066       10.10.1.46:ipp          CLOSE_WAIT
```

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4: $ netstat -r
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
default         _gateway        0.0.0.0         UG        0 0          0 enp44s0
default         _gateway        0.0.0.0         UG        0 0          0 wlo1
10.10.0.0       0.0.0.0         255.255.255.0   U         0 0          0 br-703fe8c0b04f
10.20.0.0       0.0.0.0         255.255.255.0   U         0 0          0 br-4c9471b06aa9
10.30.0.0       0.0.0.0         255.255.240.0   U         0 0          0 enp44s0
10.30.0.0       0.0.0.0         255.255.240.0   U         0 0          0 wlo1
172.17.0.0      0.0.0.0         255.255.0.0     U         0 0          0 docker0
172.18.0.0      0.0.0.0         255.255.0.0     U         0 0          0 docker_gwbridge
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4: $
```

NMAP

PURPOSE: It is an open-source tool used for network discovery and security auditing.

USAGE: nmap [options] [target]

EXAMPLE:

```
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ nmap iitbhilai.ac.in
Starting Nmap 7.80 ( https://nmap.org ) at 2026-01-05 19:54 IST
Nmap scan report for iitbhilai.ac.in (192.168.10.115)
Host is up (0.00040s latency).
Not shown: 996 closed ports
PORT     STATE SERVICE
22/tcp   open  ssh
80/tcp   open  http
443/tcp  open  https
5666/tcp open  nrpe

Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$ nmap -p 22,443 iitbhilai.ac.in
Starting Nmap 7.80 ( https://nmap.org ) at 2026-01-05 19:54 IST
Nmap scan report for iitbhilai.ac.in (192.168.10.115)
Host is up (0.00053s latency).

PORT    STATE SERVICE
22/tcp  open  ssh
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
arindam-chakraborty@arindamchakraborty-ASUS-TUF-Gaming-F15-FX507ZC4-FX577ZC4:~$
```



THANK YOU